

QUESTION OF THE WEEK 5

GRADE XI CHEMISTRY

Assignment Submission Date: Bhadra 1

1. Differentiate between electron affinity and electronegativity.
2. Compare the size of S^{2-} , Cl^- , Ar , K^+ and Ca^{++} with reason.
3. Define electron affinity. What are the various factors that affect electron affinity?
4. State and explain Graham's Law of Diffusion. If the time of equal effusion of pure oxygen and a mixture of oxygen and ozone are 224 sec and 243 sec respectively, find the percentage of ozone in the mixture.
5. State and explain Dalton's Law of partial pressure. 500 cc of oxygen under 680 mm pressure and 250 cc of nitrogen at 720 mm pressure were put together in a litre flask keeping the temperature constant. Find the final pressure of the mixture.
6. Balance the following redox reaction by ion-electron method.
 - a) $Cl_2 + OH^- \longrightarrow Cl^- + ClO_3^- + H_2O$
 - b) $MnO_4^- + H_2O_2 + H^+ \longrightarrow MnO_2 + O_2 + H_2O$

Weekly question week 5

Grade XI Physics

Assignment Submission Date: Bhadra 1

- a. If $\vec{A}$ and $\vec{B}$ are non-zero vectors, is it possible $\vec{A} \times \vec{B}$ and $\vec{A} \cdot \vec{B}$ both to be zero at the same time? Explain.
- b. What do you mean by resolution of a vector?
- c. Discuss in detail, the resolution of a vector in two dimensions.
- d. Explain the geometrical interpretation of scalar product of two vectors.
- e. Vectors; $\vec{A} = 3\hat{i} + 5\hat{j} - 2\hat{k}$ and $\vec{B} = -3\hat{j} + 6\hat{k}$. Find the vector $\vec{C}$, such that $2\vec{A} + 7\vec{B} + 4\vec{C} = 0$.
- f. Find the scalar product of the two vectors: $\vec{A} = 4.00\hat{i} + 3.00\hat{j}$ and $\vec{B} = 5.00\hat{i} + 2.00\hat{j}$. Also, find the angle between them.
- g. a. Two spheres of same radius are made up of same material but one is hollow and the other is solid. Will their overall expansion be same or different if they are heated from 20°C to 30°C ? Explain?
- h. A Steel rod and a brass rod when measured at 0°C are 50.1 cm and 50cm respectively. At what temperature will their length become equal and what will be their common length? Given, $\alpha_{\text{steel}} = 12 \times 10^{-6}/^\circ\text{C}$ and $\alpha_{\text{brass}} = 18 \times 10^{-6}/^\circ\text{C}$.
- i. Does the density of a substance depend on temperature? Explain.
- j. Derive an expression showing the variation of density of a substance with the change in temperature.
- k. The coefficient of real expansion of a substance is $1.6 \times 10^{-4}/^\circ\text{C}$. If the density of the substance is 10 g/cc at 0°C , find its density at 673K .
- l. On heating a glass block of 10,000 cm^3 , from 25°C to 40°C , its volume increases by 4cm^3 . Calculate coefficient of linear expansion of the glass.
- m. The difference between the lengths of a certain brass rod and that of a steel rod is claimed to be constant at all temperatures. Is this possible? Explain.
- n. Describe the experiment method for the determination of linear expansivity of a material in the form of a rod.
- o. A glass rod rubbed with silk is found to have acquired a positive charge of $6.4 \times 10^{-7}\text{C}$. Does the glass rod lose any mass in this process? If so, estimate the loss of mass.
- p. State and explain Coulomb's law in electrostatics.
- q. Point Charges having values $+5\mu\text{C}$, $+10\mu\text{C}$, and $-10\mu\text{C}$ are placed in corners A, B and C of an equilateral triangle ABC having each side equal to 5 cm. Find the resultant force on the charge at A.
- r. State and explain the principle of superposition of Coulomb's force.
- s. Why coulomb's law is applicable to stationary charges only?
- t. Two point charges of $10\mu\text{C}$ and $-90\mu\text{C}$ are separated by a distance of 18 cm. Find the point along the line joining them at which electric field intensity is zero.

QUESTION OF THE WEEK (WEEK 4)

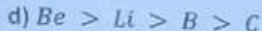
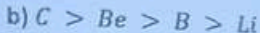
CLASS XI (sub: PHYSICS)

- 1) a) What are the limitations of dimensional analysis?
 b) Check the correctness of the physical relation $\rho = \frac{3g}{4\pi r g}$ where symbols carry their usual meanings.
 c) Find the value of 20 J on the system based on 20 cm, 1kg and $\frac{1}{2}$ minute as fundamental units.
- 2) a) State and use parallelogram law of vector addition to find the magnitude and direction of the resultant of two vectors inclined at an angle θ .
 b) Two forces F_1 and F_2 act at a point. F_1 has magnitude 8N and acts at an angle of 30° with the positive x-axis. F_2 has a magnitude of 6N and acts at an angle of 150° with positive x-axis. Find the magnitude and direction of resultant force.
- 3) a) Define the coefficient linear, superficial and cubical expansions. Derive relations between them.
 b) Why a thick glass cracks when boiling water is poured in it?
- 4) a) Pendulum clocks generally go fast in winter and slow in summer. Why?
 b) A pendulum clock loses 12 Sec. a day if the temperature is 40°C and gains 4 Sec. a day if the temperature is 20°C . Find the temperature at which the clock will show correct time and coefficient of linear expansion of the metal of pendulum shaft.
- 5) a) State and explain the principle of superposition of Coulomb's force.
 b) Two-point charges of $10\ \mu\text{C}$ and $-40\ \mu\text{C}$ are separated by distance of 13 cm. Find the point along the line joining them at which net electrostatic force on any charge is zero.
- 6) a) Define dielectric constant. Why it's value is always greater than one?
 b) Four equal charges q are placed at the four corners of a square of length a . Find the magnitude of the net force acting on one of them due to other charges and draw diagram to show the direction of force on that charge.
- 7) a) Explain why vector addition is commutative, providing a real-life example to support your reasoning.
 b) What is meant by resolution of vectors? How will you resolve a vector in a plane in two mutually perpendicular directions?
- 8) a) Why does a solid expand on heating?
 b) Explain an experiment to determine the coefficient of linear expansion of a solid.
 c) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder at 15°C is 8.0 cm and there is an all-round clearance of 0.05 mm between the piston and the cylinder of the wall. Find the temperature at which the two will exactly fit in. ($\alpha_{\text{steel}} = 1.2 \times 10^{-5}\ ^\circ\text{C}^{-1}$ and $\alpha_{\text{aluminium}} = 1.6 \times 10^{-5}\ ^\circ\text{C}^{-1}$)
- 9) a) State and explain Coulomb's law in electrostatics.
 b) Why coulomb's law is applicable to stationary charges only?
 c) Define 1 coulomb charge on the basis of Coulomb's law.
 d) Two charges 10^{-6}C and 10^{-7}C attract each other with a force of 400N. Calculate the distance between them if they are in i) air and ii) water (relative permittivity of water is 80)

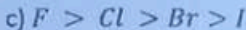
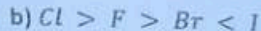
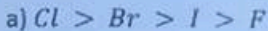
GROUP - 'A' (MCQ) [11×1=11]

- 3.7 g of a gas at 25° C occupy the same volume as 0.184 g of H_2 at 17° C and at same pressure. What is the molecular mass of gas?
 a) 40.21 amu b) 45.35 amu c) 41.33 amu d) 46.32 amu
- Diffusion of He is
 a) 4 times faster than CO_2 b) 4 times faster than SO_2
 c) 4 times faster than NO_2 d) 4 times faster than ClO_2
- The temperature below which gas does not obey ideal gas law is
 a) Critical temperature b) Inversion temperature
 c) Boyle's temperature d) Reduced temperature
- 300 ml of a gas at 27° C is cooled to - 3° C at constant pressure. The final volume of is
 a) 540 ml b) 270 ml c) 135 ml d) 350 ml
- An element M has an atomic mass 19 and the atomic number 9. It's ion is represented by
 a) M^+ b) M^{2-} c) M^- d) M^{2+}
- The oxidation number of sulphur in S_8 , S_2F_2 and H_2S are respectively
 a) 0, +1, -2 b) +2, +1, -2 c) 0, +1, +2 d) -2, +1, -2
- Of the following reactions, which involves redox process? ss
 a) $HI + AgNO_3 \longrightarrow AgI + HNO_3$
 b) $HCl + NaBr \longrightarrow NaCl + HBr$
 c) $HCl + NaOH \longrightarrow NaCl + H_2O$
 d) $H_2 + Br_2 \longrightarrow 2 HBr$

8. The correct order of decreasing first ionization energy is



9. Which of the following represents the correct order of electron affinities?



10. Which is the best reducing agent?



11. Which one of the following ions has the smallest radius?



KATHMANDU MODEL SECONDARY SCHOOL

FIRST TERM EXAMINATION 2082

Class: XI 'Science'

Time: 3hrs.

Subject: Chemistry [3011]

F.M.: 75

P.M.: 30

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

GROUP - 'B' [8x5=40]

SHORT ANSWER QUESTIONS.

- a) Define electron affinity. [1]

b) Giving reasons, explain the variation of electron affinity in a period and group. [2]

c) Be and Mg have almost zero electron affinity, why? [2]
- a) State modern periodic law. [1]

b) Discuss the various features of modern periodic table. [4]
- a) Define the terms: i) Screening Effect ii) Vander Waal's Radius [3]

iii) Isoelectronic Species [2]

b) Compare the size of Cl^- , Ar, & K^+ with reason. [2]
- a) Justify the statement, "Oxidation and Reduction reaction occurs simultaneously." [2]

b) Define oxidation and reduction in terms of oxidation number. [1]

c) Find the oxidation number of:
i) S in $H_2S_2O_8$ ii) Ag in $K[Ag(CN)_2]$ [2]
- a) Define autoredox reaction. [1]

b) Balance the following redox reaction either by oxidation number method or by ion electron method. [3]

$$Cl_2 + NaOH \longrightarrow NaCl + NaClO_3 + H_2O$$

c) Identify oxidizing agent and reducing agent. [1]
- a) State and explain Graham's law of diffusion. [2]

b) A gaseous hydrocarbon having the molecular formula C_nH_{2n+2} diffuses twice as quickly as sulphur dioxide. Determine the value of 'n' and molecular formula of hydrocarbon. [3]
- a) State and explain Dalton's law of partial pressure. [2]

- b) 500 cc of oxygen under 680 mm pressure and 250 cc of nitrogen at 720 mm pressure were put together in a litre flask keeping the temperature constant. Find the final pressure of the mixture. [3]
8. a) Differentiate between oxidation number and valency. [2]
- b) Balance the following redox reaction either by oxidation number method or by ion electron method. [3]
- $$KMnO_4 + KCl + H_2SO_4 \longrightarrow MnSO_4 + K_2SO_4 + H_2O + Cl_2$$

GROUP - 'C' [3x8=24]

LONG ANSWER QUESTIONS

9. a) Derive Ideal gas equation. [2]
- b) Find the value of R in CGS unit. [2]
- c) Define absolute zero temperature. [1]
- d) An evacuated vessel weighs 50 gm when empty, 148 gm when filled with a liquid of density of 0.98 gm/cc and 50.5 gm when filled with an ideal gas at 760 mm Hg and at 300 K. Determine the molecular mass of gas. [3]
10. a) Explain with diagram and reaction involved for the preparation of H_2S using Kipp's apparatus. [3]
- b) Write the reaction to show that H_2S acts as precipitating agent. [1]
- c) Write the principle of manufacture of sulphuric acid by contact process. Also draw the flow sheet diagram for contact process. [2+2]
11. a) Write the principle for balancing redox reaction by ion-electron method. [1]
- b) Balance the following redox reaction either by oxidation number method or by ion electron method.
- $$CrO_4^{2-} + HSnO_2^{2-} \longrightarrow CrO_2^- + HSnO_3^- \text{ (in alkaline medium)} \quad [4]$$
- c) A vessel contains 12 gm of an ideal gas at $t^\circ C$ temperature and 1 atm pressure. When the temperature is increased by $10^\circ C$ at the same volume, the pressure is increased by 10%. Calculate the volume of gas and initial temperature. [Molecular mass of gas = 120 amu]. [3]

GROUP 'A' MCQ [11×1=11]

Re-write the correct option of each question in your answer sheet.

1. If
- $x, y > 0$
- and
- $\frac{1}{x} < \frac{1}{y}$
- , which one of the following is true?

A) $x > y$ B) $-\frac{1}{x} > -\frac{1}{y}$ C) $x < -y$ D) $x < y$

2. Which one of the following is equivalent to
- $\sim(p \wedge \sim q)$
- ?

A) $\sim(p \wedge q)$ B) $(p \vee q)$ C) $(\sim p \vee q)$ D) $(\sim p \wedge q)$

3. What is the value of the
- $\lim_{x \rightarrow 1} \frac{\sin \pi x}{1-x^2}$
- ?

A) $\frac{\pi}{2}$ B) π C) $\frac{2}{\pi}$ D) Does not exist

4. If the function
- $f(x) = \begin{cases} \frac{\sin^2 ax}{x^2} & \text{for } x \neq 0 \\ 16 & \text{for } x = 0 \end{cases}$
- is continuous at
- $x = 0$
- , then what is the value of
- a
- ?

A) 16 B) 8 C) 4 D) 2

5. What is the value of
- $\lim_{t \rightarrow \infty} \frac{\sqrt{3t^2+1}}{3t-7}$
- ?

A) $\frac{7}{\sqrt{3}}$ B) $\sqrt{3}$ C) $\frac{1}{\sqrt{3}}$ D) $\frac{-1}{\sqrt{3}}$

6. What is the value of
- $\lim_{y \rightarrow \pi} \frac{\sin y}{2+\cos y}$
- ?

A) 0 B) 1 C) π D) 2

7. If
- $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$
- and
- $\mathbb{R}$
- have their usual meaning, then which one of the following statements is not true?

A) $\mathbb{Z} \cap \mathbb{Q} = \mathbb{Z}$ B) $\mathbb{Z} \subset \mathbb{Q}$ C) $\mathbb{R} \subset \mathbb{Z}$ D) $\mathbb{Q} \cup \mathbb{R} = \mathbb{R}$

8. What is the value of $\lim_{x \rightarrow 0^+} (\frac{1}{x} - \frac{1}{|x|})$?

A) -1

B) 0

C) ∞

D) 1

9. What is the value of $\frac{\Delta y}{\Delta x}$, if $y = 2 - 3x$?

A) 2

B) -3

C) 5

D) 3

10. If $A = \{1, 2, 3\}$, what is the value of $n[P(A)]$?

A) 4

B) 15

C) 16

D) 8

11. What does $x \notin (B - A)$ mean?

A) $x \in A$ and $x \notin B$

C) $x \in B$ and $x \notin A$

B) $x \in A$ or $x \notin B$

D) $x \notin A$ or $x \in B$

Class: XI 'Science'

Time: 3 hrs.

Subject: Com. Mathematics (0071)

*Candidates are required to give their answers in their own words as far as practicable.
 The figures in the margin indicate full marks.*

Attempt all the questions.

GROUP 'B' [8×5=40]

- | | | |
|--------|--|-------|
| 12. a) | Define symmetric difference of two sets. | [1] |
| b) | For what value of x , $\sqrt{x}$ is real a number? | [1] |
| c) | What are conjugates of conditional $p \Rightarrow q$? | [1] |
| d) | Define a closed interval. | [1] |
| e) | Define the contradiction of a statement. | [1] |
| | | |
| 13. a) | If $P \subseteq Q$, then show that $\bar{Q} \subseteq \bar{P}$. | [2] |
| b) | For any three sets X, Y and Z , show that
$(X \cup Y) - Z = (X - Z) \cup (Y - Z)$ | [3] |
| | | |
| 14. a) | Write the antecedent and the consequent of the statement
<i>"if 5 is a natural number then $\frac{1}{5}$ is a rational number."</i> | [2] |
| b) | Let p, q and r be any statements. Show that the following
statement is true. $(p \Rightarrow q) \wedge (q \Rightarrow r) \Rightarrow (p \Rightarrow r)$ | [3] |
| | | |
| 15. a) | Write any two properties of inequalities. | [2] |
| b) | Solve the inequality: $-3 < 7 - 2x \leq 5$.
Also, represent the solution set on the number line. | [2+1] |
| | | |
| 16. a) | For any two sets A and B , prove that
$\overline{(A \cap B)} = \bar{A} \cup \bar{B}$ | [2] |
| b) | If $A = [-1, 3]$ and $B = (0, 2)$ then find $A \cup B, A \cap B$ and $A - B$ | [3] |
| | | |
| 17. a) | Let $h(x)$ be a continuous function. What is the relationship
between $\lim_{x \rightarrow p} h(x)$ and $h(p)$? | [1] |
| b) | Define derivative of $f(x)$ with respect to x . | [1] |
| c) | If $\lim_{x \rightarrow c} f(x)$ exists, what is the relationship between
$\lim_{x \rightarrow c^-} f(x)$ and $\lim_{x \rightarrow c^+} f(x)$? | [1] |
| d) | Define jump discontinuity of a function $f(x)$ at $x = a$. | [1] |
| e) | What is the value of $\lim_{x \rightarrow 0} \frac{e^{\sqrt{2}x} - 1}{x}$? | [1] |

18. a) Evaluate the limit, if it exists. $\lim_{x \rightarrow 16} \frac{4 - \sqrt{x}}{16x - x^2}$ [2]
 b) If $f(x) = \begin{cases} \frac{\sqrt{1+px} - \sqrt{1-px}}{x}, & -1 \leq x < 0 \\ \frac{2x+1}{x-2}, & 0 \leq x \leq 1 \end{cases}$ [3]
 is continuous at $x = 0$. Find the value of p .

19. a) Find the limit of the following function if it exists: [2]
 $f(x) = \begin{cases} 2-x & \text{for } x < 2 \\ 2+x & \text{for } x \geq 2 \end{cases}$ at $x = 2$.
 b) i) Evaluate: $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x}$ [2]
 ii) Is it possible to write $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = \lim_{x \rightarrow 0} x^2 \times \lim_{x \rightarrow 0} \sin \frac{1}{x}$? If not, give reason. [1]

GROUP 'C' (3×8=24)

20. a) For any two statements p and q , prove that $\sim(p \vee q) \equiv \sim p \wedge \sim q$. [2]
 b) Define a **Statement**. How does it differ from **sentence** in the context of logic? Explain. [2]
 c) What is the relationship between the **existence of limits** and the **continuity of a function** at a particular point? Explain. [4]

21. a) Let $f(x) = \begin{cases} \frac{\tan x - \cot x}{x - \frac{\pi}{4}}, & x \neq \frac{\pi}{4} \\ a, & x = \frac{\pi}{4} \end{cases}$ be continuous at $x = \frac{\pi}{4}$. Find [4]
 the value of a .
 b) Find the derivative of the function [4]
 $f(x) = \sqrt[3]{(2-3x)^2}$ by using definition of the derivative.

22. a) Evaluate: $\lim_{x \rightarrow y} \frac{\tan x - \tan y}{x - y}$ [2]
 b) Find from the first principles, the differential coefficient of the [3]
 function $\cot(b - ax)$.
 c) If A and B be any two non-empty sets, verify that
 $A - B$ and $B - A$ are disjoint sets by choosing suitable sets A and B . [3]



Kathmandu Model Secondary School
KATHMANDU MODEL SECONDARY SCHOOL

FIRST TERM EXAMINATION 2082

Class: XI 'Science'

SET 'B'

F.M.: 50

Time: 2 hrs.

Subject: Computer Science (4271)

P.M.: 20

Candidates are required to give their answer in their own words as far as practicable.

The figure in the margin indicated full marks:

Attempt all the questions.

GROUP 'A' [9×1=9]

MULTIPLE CHOICE QUESTIONS.

- The three main components of computer are
 - CPU, Memory, Tape
 - I/O, Printer, Mouse
 - Tape, I/O, Floppy disk
 - CPU, I/O, Memory
- Which type of computer is smaller than a mainframe but larger than a microcomputer, and used by medium-sized organizations?
 - Hybrid Computers
 - Super Computers
 - Mini Computers
 - Analog Computers
- Which is the major component of fourth generation computers?
 - Transistor
 - Microprocessor
 - Vacuum Tube
 - IC
- A speaker is an example of device.
 - processing
 - storage
 - input
 - output
- What is the binary equivalent of a decimal number 31?
 - 10101
 - 11111
 - 11010
 - 101101
- How many bits are there in 2.5 kilobytes?
 - 20,480 bits
 - 25,600 bits
 - 40,960 bits
 - 20,480 bytes
- The maximum number of bits used to represent hexadecimal number in binary is
 - 3
 - 4
 - 8
 - 16
- The _____ is a small input device used to select and display objects on a screen.
 - Ink
 - Magnet
 - Light pen
 - None of these
- Which of the following is not the font style?
 - Bold
 - Italics
 - Regular
 - Superscript

1024 byte